

From Predictive Entropy to Semantic Entropy: A Survey of Uncertainty Metrics for Large Language Models

Abstract

Reliable uncertainty estimation is a critical requirement for deploying Large Language Models (LLMs) in real-world applications. Traditional uncertainty metrics primarily rely on token probabilities and predictive entropy, which often fail to capture semantic equivalence in free-form text generation [5, 4]. This survey reviews the evolution of uncertainty quantification metrics in LLMs, focusing on the transition from token-level probability measures to semantic-level approaches such as Semantic Entropy [4, 2]. The survey compares the theoretical foundations, mathematical formulations, and empirical performance of existing uncertainty metrics [8, 9]. Additionally, it discusses the challenges of evaluating uncertainty in open-ended generation tasks and identifies key directions for future research in semantic-aware uncertainty estimation.

1 Introduction

Large Language Models (LLMs) have become central components in modern natural language processing systems, including question answering, summarization, code generation, and decision-support applications. Despite their strong generative capacity, LLMs often produce plausible but factually incorrect outputs, commonly described as hallucinations or confabulations [4, 2, 9]. This limitation is particularly problematic in high-stakes domains such as medicine, law, education, and scientific research, where model outputs may influence consequential decisions [2, 9]. Therefore, uncertainty quantification (UQ) is not merely an auxiliary diagnostic tool; it is a mechanism for assessing whether a generated response should be trusted, verified, or deferred to human review [4, 2].

Earlier approaches to uncertainty estimation in language generation were largely based on token probabilities and predictive entropy [5]. These methods inherit the probabilistic

structure of autoregressive language modeling, where a sequence is evaluated through the product of conditional token probabilities. However, free-form text generation introduces a structural difficulty: the same semantic answer can be expressed through many different lexical and syntactic forms [4]. As a result, token-level entropy may assign high uncertainty to a model that is semantically confident but lexically diverse [4, 2]. This mismatch has motivated a shift from surface-form uncertainty to semantic-level uncertainty.

This survey summarizes this transition from predictive entropy to semantic entropy. It first introduces the major types of uncertainty in LLMs, then reviews token-level predictive entropy and its limitations. Next, it explains Semantic Entropy and related semantic-aware extensions, including Discrete Semantic Entropy and Kernel Language Entropy. Finally, it discusses empirical trends, operational constraints, and future directions for uncertainty estimation in open-ended generation tasks [2, 7, 8, 9].

2 Uncertainty in Language Models

Uncertainty in machine learning is commonly divided into aleatoric and epistemic uncertainty [9]. Aleatoric uncertainty refers to irreducible ambiguity in the data or task. In language generation, this occurs when a prompt admits multiple valid continuations. For example, a prompt asking for a greeting may validly produce “Hello,” “Hi,” or “Good morning.” High uncertainty in this case reflects linguistic variability rather than model ignorance [1, 9].

Epistemic uncertainty, by contrast, refers to uncertainty caused by insufficient model knowledge [9]. It arises when the model lacks reliable information about the queried fact, domain, or reasoning process. Hallucination is more closely associated with epistemic uncertainty than with aleatoric uncertainty, because the model may generate a fluent answer despite lacking sufficient factual grounding [2, 9]. The central goal of modern UQ methods is therefore to distinguish benign variation in expression from genuine uncertainty about meaning or truth.

Existing UQ methods for LLMs can be broadly classified into four groups: verbalized confidence methods, latent-state methods, token- or sequence-probability methods, and semantic clustering methods [3, 5, 6, 9]. Table 1 summarizes this classification.

Among these categories, semantic clustering methods are especially important for free-form generation because they evaluate uncertainty at the level of meaning rather than at the level of exact token sequences [4, 2].

Table 1: Major categories of uncertainty estimation methods for LLMs, adapted from recent taxonomies of LLM uncertainty estimation [9].

Category		Core mechanism	Examples
Verbalized confidence	confidence	Prompts the model to report or judge its own confidence.	P(True), confidence prompting [3]
Latent-state methods	meth-ods	Uses hidden states, activations, or internal representations.	Latent-state classifiers, SE-LSC [9]
Token/sequence probability methods		Measures uncertainty from token probabilities or sequence likelihoods.	Predictive Entropy, length-normalized entropy [5]
Semantic clustering methods	clustering	Groups multiple outputs by meaning and estimates uncertainty over semantic clusters.	Semantic Entropy, DSE, KLE [4, 2, 7]

3 Predictive Entropy and Its Limitations

In autoregressive language modeling, a generated sequence $s = (t_1, t_2, \dots, t_N)$ is assigned a probability conditioned on an input prompt x [5]:

$$p(s|x) = \prod_{i=1}^N p(t_i | t_{<i}, x). \quad (1)$$

Predictive entropy estimates the uncertainty of the model’s output distribution. Given M sampled sequences, a common Monte Carlo approximation is [5]:

$$H(x) \approx -\frac{1}{M} \sum_{m=1}^M \log p(s_m|x). \quad (2)$$

For categorical answer tokens, entropy can also be written in the Shannon form:

$$H = -\sum_{i=1}^N p(a_i|x) \log p(a_i|x). \quad (3)$$

These formulations are mathematically straightforward, but they have two important weaknesses in language generation. First, sequence probability decreases with length because it is a product of probabilities less than or equal to one [5]. Longer responses can therefore appear less certain even when they are correct. Length normalization partly mitigates this issue by using average token confidence, but it remains a heuristic adjustment rather than a complete solution [4, 5].

Second, predictive entropy is sensitive to lexical variation. Suppose a model answers the question “What is the capital of France?” with several paraphrases: “Paris,” “The capital is Paris,” and “It is Paris.” Token-level entropy treats these as different sequences, even though they express the same fact. The probability mass is divided across surface forms, which can artificially increase entropy [4, 2]. This problem is often discussed as the failure of token-level uncertainty to respect linguistic invariance: the model is semantically certain, but token-level metrics interpret lexical diversity as uncertainty [4, 6].

4 Semantic Entropy

Semantic Entropy addresses the above limitation by estimating uncertainty over meanings rather than over surface forms [4, 2]. The key assumption is that multiple generated sequences may belong to the same semantic equivalence class. Let $\mathcal{C}$ denote a set of semantic clusters, where each cluster $C \in \mathcal{C}$ contains sequences with equivalent meaning. The probability of a semantic cluster is obtained by aggregating the probabilities of sequences in that cluster [4]:

$$p'(C|x) = \sum_{s \in C} p'(s|x). \quad (4)$$

Because only a finite set of generations is sampled, cluster probabilities are often renormalized over observed clusters:

$$p''(C|x) = \frac{p'(C|x)}{\sum_{C \in \mathcal{C}} p'(C|x)}. \quad (5)$$

Semantic Entropy is then computed as the Shannon entropy over semantic clusters [4, 2]:

$$SE(x) \approx - \sum_{C \in \mathcal{C}} p''(C|x) \log p''(C|x). \quad (6)$$

This formulation resolves the probability dilution problem. If all sampled answers express the same meaning, their probability mass is aggregated into one dominant cluster, and semantic entropy becomes low. If the model generates mutually incompatible answers, probability mass is distributed across several semantic clusters, and semantic entropy becomes high [4, 2]. Thus, Semantic Entropy is better aligned with epistemic uncertainty in factual generation than raw token entropy.

In practice, semantic clusters are usually constructed using Natural Language Inference

(NLI) models [4, 2]. Two generations are placed in the same cluster if they entail each other bidirectionally. This bidirectional entailment criterion approximates semantic equivalence: if answer s_A entails s_B and s_B entails s_A , the two answers are treated as expressing the same meaning [4]. This method is effective for factual question answering, but it introduces additional computational cost and depends on the reliability of the external NLI classifier [2, 9].

5 Extensions of Semantic Uncertainty Metrics

Several extensions have been proposed to reduce the operational limitations of Semantic Entropy or to generalize it beyond hard clustering [2, 7, 9].

5.1 Discrete Semantic Entropy

Standard Semantic Entropy requires sequence likelihoods, which are available only when token-level probabilities or logits are accessible [2, 8]. This makes it difficult to apply to black-box LLM APIs. Discrete Semantic Entropy (DSE) replaces sequence likelihoods with empirical cluster frequencies [2]. Given n sampled outputs and k semantic clusters, DSE is defined as

$$DSE(q|\theta) = - \sum_{i=1}^k \left(\frac{1}{n} \sum_d \mathbb{1}_{d \in c_i} \right) \log \left(\frac{1}{n} \sum_d \mathbb{1}_{d \in c_i} \right). \quad (7)$$

DSE is compatible with black-box models because it only requires sampled outputs [2]. However, it may underestimate true uncertainty when the sample size is small, since low-probability semantic alternatives may not be observed. This is related to the unseen-class problem in finite-sample estimation [2].

5.2 Kernel Language Entropy

Semantic Entropy and DSE rely on hard cluster assignments, which can be too rigid for long-form generation [7, 9]. Kernel Language Entropy (KLE) instead represents semantic relatedness through a continuous kernel matrix [7]. Pairwise semantic similarities among generated outputs are encoded in a positive semidefinite matrix K , and uncertainty is measured using von Neumann entropy:

Table 2: Comparison of representative uncertainty estimation methods based on their main operational trade-offs [3, 5, 4, 2, 7, 9].

Method	Main advantage	Main limitation
Predictive Entropy	Simple and directly based on token probabilities.	Confounds lexical diversity with semantic uncertainty.
Verbalized confidence	Easy to apply in black-box settings.	Depends on model self-calibration and may be overconfident.
Semantic Entropy	Measures uncertainty over meanings rather than token strings.	Requires multiple samples and external semantic clustering.
Discrete Semantic Entropy	Black-box-compatible semantic uncertainty metric.	Sensitive to finite-sample underestimation.
Kernel Language Entropy	Captures soft semantic similarity among outputs.	Requires construction and decomposition of a kernel matrix.

$$KLE(x) = -\text{Tr}(K \log K) = -\sum_{j=1}^N \lambda_j \log \lambda_j, \quad (8)$$

where λ_j denotes the eigenvalues of K [7]. If the outputs are semantically concentrated, one eigenvalue dominates and entropy is low. If the outputs span multiple semantic directions, the eigenvalues become more dispersed and entropy increases. KLE therefore provides a softer formulation of semantic uncertainty and can better handle partial semantic overlap [7].

6 Empirical Evaluation and Practical Constraints

Uncertainty metrics are commonly evaluated by their ability to distinguish correct from incorrect generations. Standard metrics include AUROC and selective prediction curves such as the area under the accuracy-rejection curve [5, 9]. Empirical studies generally find that semantic-level metrics outperform token-level predictive entropy on free-form factual question answering tasks, particularly when correct answers can be expressed in multiple ways [4, 2, 8].

The primary advantage of semantic methods is their robustness to paraphrase variation.

On datasets such as TriviaQA and CoQA, semantic entropy is often more informative than predictive entropy because it evaluates whether sampled responses converge on the same meaning [4, 2, 8]. However, semantic methods also introduce practical constraints. First, they require multiple generations per prompt, increasing inference cost [8]. Second, NLI-based clustering has quadratic pairwise complexity with respect to the number of sampled outputs [7]. Third, the quality of the uncertainty estimate depends on the semantic evaluator; an NLI model trained on general-domain data may fail in specialized domains such as medicine, law, or theology [4, 9].

7 Future Directions

Future research on LLM uncertainty estimation is likely to focus on three directions. First, compute-efficient semantic uncertainty is needed. Instead of using a fixed number of samples for every prompt, adaptive sampling methods can stop generation once the semantic distribution stabilizes. This would allocate more computation to genuinely uncertain prompts and less computation to simple prompts [1, 8].

Second, stronger black-box estimators are necessary because many widely used LLMs expose limited or no token-probability information [2, 8]. Frequency-based semantic estimators such as DSE are promising, but they require better correction for unobserved semantic classes under small sample budgets [2].

Third, semantic uncertainty must be extended beyond short factual answers. Long-form generation, reasoning chains, multimodal outputs, and domain-specific language all involve partial entailment and nuanced semantic relations [7, 9]. Continuous approaches such as KLE and latent-space semantic clustering may provide more flexible tools for these settings [7, 9].

8 Conclusion

The development of uncertainty estimation for LLMs has moved from token-level probability measures toward semantic-level uncertainty metrics. Predictive entropy provides a useful probabilistic baseline, but it is poorly matched to free-form generation because it treats paraphrases as separate outcomes [5, 4]. Semantic Entropy corrects this limitation by aggregating probability mass over equivalence classes of meaning, thereby better separating lexical variation from epistemic uncertainty [4, 2]. Extensions such as Discrete Semantic

Entropy and Kernel Language Entropy further broaden the applicability of semantic-aware uncertainty estimation to black-box models and continuous semantic spaces [2, 7]. Although computational cost and evaluator reliability remain major barriers, semantic uncertainty metrics provide a more appropriate foundation for assessing the reliability of LLM outputs in open-ended generation tasks [8, 9].

References

- [1] Duan, J., Cheng, H., Wang, S., Zavalny, A., Wang, C., Xu, R., Kailkhura, B., & Xu, K. (2024). Shifting attention to relevance: Towards the predictive uncertainty quantification of free-form large language models. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.
- [2] Farquhar, S., Kossen, J., Kuhn, L., & Gal, Y. (2024). Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017), 625–630.
- [3] Kadavath, S., Conerly, T., Askell, A., Henighan, T., Drain, D., Perez, E., et al. (2022). Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*.
- [4] Kuhn, L., Gal, Y., & Farquhar, S. (2023). Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. *Proceedings of the Eleventh International Conference on Learning Representations*.
- [5] Malinin, A., & Gales, M. (2021). Uncertainty estimation in autoregressive structured prediction. *Proceedings of the International Conference on Learning Representations*.
- [6] Manakul, P., Liusie, A., & Gales, M. J. F. (2023). SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. *Proceedings of EMNLP*, 9004–9017.
- [7] Nikitin, A., Kossen, J., Gal, Y., & Marttinen, P. (2024). Kernel language entropy: Fine-grained uncertainty quantification for LLMs from semantic similarities. *Advances in Neural Information Processing Systems*.
- [8] Vashurin, R., Fadeeva, E., Vazhentsev, A., Rvanova, L., Vasilev, D., Tsvigun, A., et al. (2025). Benchmarking uncertainty quantification methods for large language models with LM-Polygraph. *Transactions of the Association for Computational Linguistics*, 13, 220–248.

- [9] Xia, Z., Xu, J., Zhang, Y., & Liu, H. (2025). A survey of uncertainty estimation methods on large language models. *Findings of the Association for Computational Linguistics: ACL 2025*, 21381–21396.